# Boolean conditions

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Python readiness portfolio

THE CONNECTED WEEK

## Teach conditions through explicit contracts and counterexamples.

Supply valid-type toy inputs, preserve each sub-result and assess a fresh multi-requirement script.

**Keep one habit:** Ask which exact requirement failed and which input would distinguish the two proposed rules.

MAIN LESSON · SECTION 1

## Prepare the connected bridge lesson

> Check prior exposure and keep execution practice distinct from paper reasoning.

### Goal and main plan

Primary target: Translate stated requirements into **comparison**s

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

### Retrieval answer guidance

## Retrieval and prerequisite guidance

Without opening the old lesson, explain one key idea from Expressions and operator order. Give an example and a mistake that the idea helps you avoid.

Look for: Evaluate numeric expressions using Python’s grouping rules

Reference example, not the only acceptable wording: 3 * 8 + 2 or (3 * 8) + 2 gives 26. 3 * (8 + 2) gives 30 and adds two to each of three packs, not two once.

### From a paper algorithm to Python

Without opening the old lesson, explain one key idea from From a paper algorithm to Python. Give an example and a mistake that the idea helps you avoid.

Look for: Translate a short paper calculation into a Python script

Reference example, not the only acceptable wording: packs = 4; per_pack = 6; total = packs * per_pack; print(total), each on its own line. The first two bind quantities, the third computes 24 and the fourth displays 24.

Assignment alone produces no displayed **output** in this script.

Use fictional activities and **Boolean** operands throughout. Do not equate suitability with permission or claim that a programmed condition learns from data.

Reference for teacher precision: https://docs.python.org/3/library/stdtypes.html#**boolean**-operations-and-or-not. Python and/or may return operands of other types; this lesson deliberately confines them to **Boolean**s.

## Materials and readiness

Python 3 editor and conditions.py

True and False cards and written activity requirements

Blank **input**, separate checks, expected and observed columns

False is a **Boolean** value; "False" is text. This week uses actual **Boolean** values for yes-or-no flags.

## End goal

Trace **boolean** conditions and explain the changed program before running it.

▼ Earlier wording, alternatives and picture transcripts

**Teach the learning cycle**

**Expressions and operator order**

**Outcomes**

Translate stated requirements into **comparison**s

Combine **Boolean** values with and or and not

Choose a case that distinguishes two conditions

**Preparation**

**Prerequisite answer**

**End goal for the pictorial case**

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

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MAIN LESSON · SECTION 2

# Teach partial checks before the combined decision

> Use distinguishing cases and keep both toy contracts labelled.

## Read minutes/quiet requirements and syntax

A fictional activity must take at most **12 minutes** and be quiet. An activity taking exactly **12 minutes** meets the time requirement. A quiet 15-minute activity does not.

Before writing code, underline each requirement and decide whether all requirements must hold or whether alternatives are allowed.

Use a fresh conditions.py file with nonnegative integer minutes and an actual **Boolean** quiet flag. A **comparison** produces True or False for these values. The expression minutes <= 12 includes 12; minutes < 12 excludes it.

Use == for equality and != for inequality. A single = assigns a value to a name; it does not ask whether two values are equal.

The script prints True then False. The first result concerns time only. The second requires time_ok and quiet. These are separate claims.

Trace each **input** before explaining the combined result; a result without its conditions can be misleading.

## Use the Boolean-only truth table

| A | B | A and B | A or B |
|---|---|---|---|
| False | False | False | False |
| False | True | False | True |
| True | False | False | True |
| True | True | True | True |

With **Boolean** operands, and needs both parts True. **Inclusive or** needs at least one True, so both True is allowed. not reverses a **Boolean**: not True is False and not False is True.

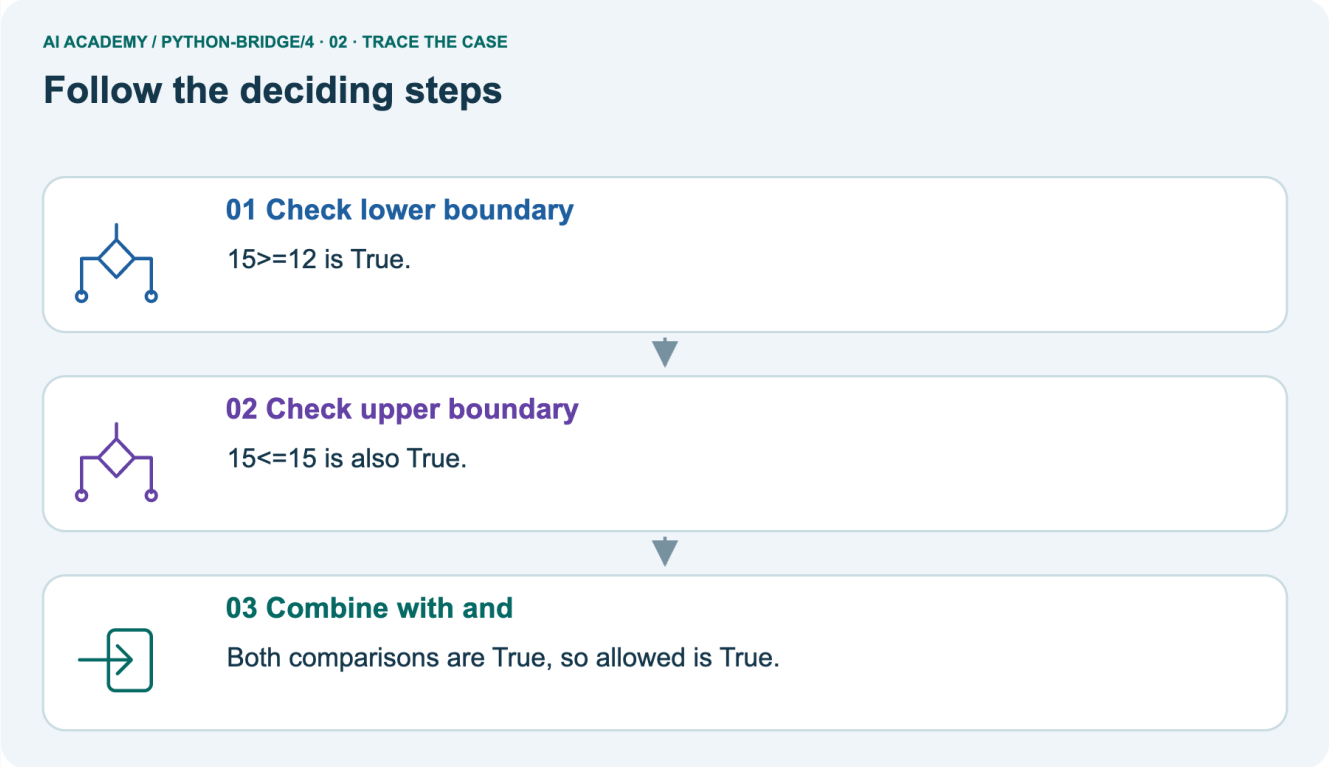
Python can also apply and and or to other types; keep operands **Boolean** here so this table directly describes the result.

## Read the separate age interval

Two conditions for entry. A toy club accepts integer ages from 12 through 15 inclusive.

Python 3: age=15; allowed=(age>=12) and (age<=15); print(allowed). No type validation is implemented here.

## Trace lower, upper and combined results




Follow the labelled picture with the matching explanation. Open the vector diagram

### Read the picture one step at a time

| STEP | WHAT HAPPENS AND WHY |
| --- | --- |
| 1. Check lower boundary | 15>=12 is True. |
| 2. Check upper boundary | 15<=15 is also True. |
| 3. Combine with and | Both comparisons are True, so allowed is True. |

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

A **Boolean** condition represents a yes-or-no result. And requires both conditions; checking only one boundary would accept values outside the intended interval.

```
age = 15
lower_ok = age >= 12
upper_ok = age <= 15
allowed = lower_ok and upper_ok
print(lower_ok)
print(upper_ok)
print(allowed)
```

| Supplied integer age | age>=12 | age<=15 | Both requirements |
| --- | --- | --- | --- |
| 11 | False | True | False |
| 12 | True | True | True |
| 15 | True | True | True |
| 16 | True | False | False |

## Predict the upper-outside case

Change: Change age to 16.

**Prediction**: The result becomes False.

Reason: The lower condition remains True, but the upper condition fails.

## State supplied-type and authority limits

This condition checks a declared numeric interval, not identity or authorisation; unexpected **input** types need separate handling.

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

### ▼ Additional guided counterexamples

**Inclusive or** accepts both available; exactly one would reject both. Exit: (12, True) should pass and reveals <= changed to <. (8, False) should fail and reveals and changed to or.

Other **input**s are acceptable if they distinguish the versions.

### ▼ Earlier wording, alternatives and picture transcripts

| Term | Meaning |
| --- | --- |
| Comparison | An expression that checks a relation between values |
| Boolean | A type with the values True and False |
| Conjunction | A requirement that both conditions hold |
| Inclusive or | A condition satisfied when either or both parts hold |
| Counterexample | A case that shows a claim or rule is wrong |

### Two requirements must both hold

| INPUT CASE | TIME CHECK | COMBINED RESULT |
|---|---|---|
| 12 min, quiet False | 12 <= 12 is True | True and False is False |
| 12 min, quiet True | 12 <= 12 is True | True and True is True |
| 15 min, quiet True | 15 <= 12 is False | False and True is False |

Passing only one requirement does not make the activity suitable.

## Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

> **A game requires a pencil or a crayon. If both are available, does inclusive or accept the case? Explain how this differs from requiring exactly one.**

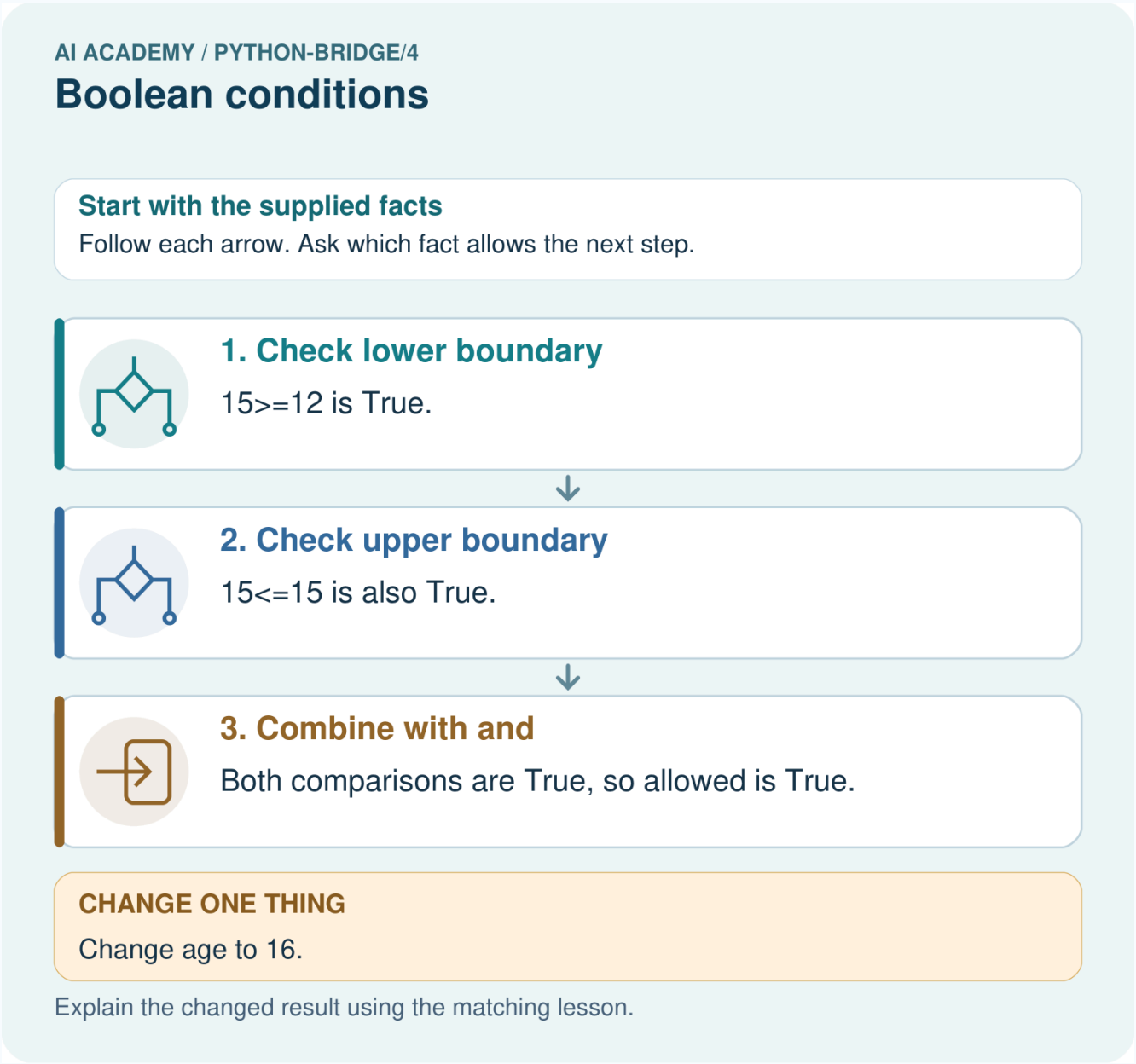
| Thinking move | Teacher prompt |
|---|---|
| Plan | What is given? What is the question? Which procedure fits these facts? |
| Do | Point to each input and intermediate step in the worked trace. Name the rule that allows the next step. |

| Thinking move | Teacher prompt |
| --- | --- |
| Check | Does the output answer the question? Which assumption or limitation prevents a larger claim? |

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

## A pictorial worked case: Boolean conditions

### The facts we will use




Follow the labelled picture with the matching explanation. Open the vector diagram

### Why this result follows

### What this example does not establish

### Change one thing in the pictured case

### Ask while showing the picture

MAIN LESSON · SECTION 3

# Guide the trace, then fade the help

> Collect predictions before execution or feedback.

## Main completion task and key

Completion task: For (minutes, quiet) cases (11, True), (12, False), and (13, True), write the time **comparison** and final suitability values before running the program.

Teacher check: Time results are True, True, False. Combined results are True, False, False; the second case fails quiet and the third fails time.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

## Individual misconception check

Use the worked case for **Boolean** conditions. Proposed explanation: "At most excludes the boundary". Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: At most 12 includes 12; compare < and <= at exactly 12.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

## Pictorial question and changed-case key

Give both **comparison** results for ages 11, 15 and 16. Which **comparison** rejects each outside value, and why does and matter?

Age 11 gives False/True, so False. Age 15 gives True/True, so True. Age 16 gives True/False, so False. Both interval boundaries must hold.

Changed-case reasoning: The result becomes False. The lower condition remains True, but the upper condition fails.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

### ▼ Support and alternative plan

| Time or stage | Teacher action |
|---|---|
| Session 1 start to 7 minutes | Check Boolean versus text with the prerequisite. |
| 7 to 20 minutes | Underline the two requirements and complete W1. |
| 20 to 33 minutes | Model the diagram and trace the two printed values. |
| 33 to 45 minutes | Complete W2 with predictions before running. |
| Session 2 start to 8 minutes | Retrieve an inclusive boundary and a one-condition failure. |
| 8 to 20 minutes | Build the four-row truth table and complete W3 and the guided question. |
| 20 to 35 minutes | Collect W4 independently, including a counterexample. |
| 35 to 45 minutes | Discuss W5 and use the exit to diagnose boundary and logic errors. |

Use the worked example in Sections 3 and 6. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Use separate cards for each **comparison**, then combine them. Read at most and at least aloud with number lines.

Extension: explain why not ((minutes >= 5) and (minutes <= 9)) identifies out-of-range numeric values under the stated **input** contract.

▼ **Earlier wording, alternatives and picture transcripts**

Compact alternative: this two-session sequence covers selected core tasks. Use the three-session weekly plan for the full lesson, independent transfer, feedback and project work; allow extra time for extensions.

**Reduce help deliberately**

**Individual reasoning check**

**Pictorial practice question**

**Reasoned teacher answer**

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MAIN LESSON · SECTION 4

# Run the complete supplied investigation

> Preserve expected, observed and paper-trace records.

## Supplied case materials and checked guidance

W2 time values True, True, False and suitability True,False,False. Boundary (12,True) distinguishes <= from <; (8,False) distinguishes AND from OR. Toy age 11→False, 12→True, 15→True, 16→False; no type validation is claimed. Selected original script outputs True,False,True,True; genre history gives False,False,True,True. Full eligibility fails availability; OR cannot substitute for a mandatory science condition.

Earlier W4 has its own case.

## Diagnose misconceptions

At most excludes the boundary: At most 12 includes 12; compare < and <= at exactly 12.

or means exactly one: **Inclusive or** also accepts two True values.

A single equals sign tests equality: = assigns; == compares.

A True result proves a fact about the world: It establishes only that the supplied values satisfy this expression.

Use Section 9 for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

## Choose accessible support

| What the response shows | Next teaching action |
|---|---|
| Missing or misread facts | Read the case aloud, mark supplied versus unknown facts, and let the learner restate it. |
| Wrong procedure | Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation. |
| Arithmetic or implementation error | Trace intermediate values or lines; preserve the first mismatch and retry that step. |
| Correct result without a reason | Ask for a trace and a changed case before marking independent understanding. |
| An unsupported wider claim | Ask which supplied evidence supports the claim and which further observation would be needed. |

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

**Match the support to the first error**

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MAIN LESSON · SECTION 5

## Assess the selected bridge transfer task

> If a learner has seen this exact case, use a fresh parallel input and record exposure/support.

### Selected independent task and checked key

Selected case: Predict all four **boolean**s. Eligibility requires science AND at most100 pages AND availability. Repeat with genre="history" and all other values unchanged.

Which OR result could wrongly suggest the genre requirement passed?

```
pages = 90
genre = "science"
available = False
print(genre == "science" and pages <= 100)
print((genre == "science" and pages <= 100) and available)
print(not available)
print(genre == "science" or pages <= 100)
```

Original results: True, False, True, True. Full eligibility is False because availability is False. With history: False, False, True, True.

The final OR remains True because90<=100; it does not establish that the science requirement passed. NOT available is True in both traces.

## Judge trace and support

| Dimension | Evidence required |
| --- | --- |
| Explanation | Names the applicable idea or procedure and ties it to the supplied facts. |
| Trace and result | Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution. |
| Limits | States the assumptions, missing evidence or conditions under which the result would change. |

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

## Give one repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

## W1 Translate the requirement

Write a condition for an activity taking at most **12 minutes** and being quiet. Explain why exactly **12 minutes** is included.

(minutes <= 12) and quiet. At most includes equality, so <= is required; the nonnegative integer and **Boolean input** contract still applies.

## W2 Trace separate checks

For (minutes, quiet) cases (11, True), (12, False), and (13, True), write the time **comparison** and final suitability values before running the program.

Time results are True, True, False. Combined results are True, False, False; the second case fails quiet and the third fails time.

## W3 Distinguish alternatives

Complete all four **Boolean** combinations for pencil or crayon. Explain the both-available case and the effect of not pencil.

or gives False, True, True, True for FF, FT, TF, TT. Both available is accepted. not pencil is True when pencil is False and False when pencil is True.

## W5 Separate a check from a guarantee

Explain why a True suitability result does not prove that the supplied duration is accurate or that a child has permission to start the activity.

The expression checks supplied values against stated conditions. Inaccurate **input** can still pass. Permission is a separate **human decision** and must not be inferred from suitability.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

### Guided and exit questions

**Inclusive or** accepts both available; exactly one would reject both. Exit: (12, True) should pass and reveals <= changed to <. (8, False) should fail and reveals and changed to or.

Other **input**s are acceptable if they distinguish the versions.

### W1 Translate the requirement

### W2 Trace separate checks

### W3 Distinguish alternatives

### W4 Write a new rule independently

A fictional puzzle qualifies when its duration is at least 5 and at most 9 minutes and its available flag is True. Write the condition. Predict results for (5, True), (9, True), (10, True), and (7, False).

Choose a case that detects replacing and with or between the range check and available.

(minutes >= 5) and (minutes <= 9) and available. Results True, True, False, False. (7, False) distinguishes (range_check and available) from (range_check or available): correct False versus faulty True.
</details>

**W5 Separate a check from a guarantee**

**Independent assessment and fresh retry**

W4 requires both inclusive bounds, the available flag, all four outcomes and a distinguishing case with reasoning.

After support, retry a 2-to-4-minute inclusive rule with an open flag: (2, True) passes, (4, False) fails and (5, True) fails. Record initial independent work before the retry.

**Assess independent transfer**

**Selected case answer**

**Feedback that the learner uses**

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MAIN LESSON · SECTION 6

# Keep the portfolio, support and delayed check

> Track code execution separately from a correct paper trace.

## Project evidence

Add a labelled **boolean** conditions **evidence** sheet to Python readiness portfolio. Show how you can translate stated requirements into **comparison**s.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

## Support and selected assessment guidance

Use separate cards for each **comparison**, then combine them. Read at most and at least aloud with number lines.

Extension: explain why not ((minutes >= 5) and (minutes <= 9)) identifies out-of-range numeric values under the stated **input** contract.

Selected independent answer: Original results: True, False, True, True. Full eligibility is False because availability is False. With history: False, False, True, True. The final OR remains True because90<=100; it does not establish that the science requirement passed. NOT available is True in both traces. Keep earlier W4 transfer as additional practice; do not substitute its numbers or key for this selected task.

Record prior exposure and use a fresh parallel case after support.

## Delayed return

Suggested return points in this level: python-bridge/5, python-bridge/7, python-bridge/10. Use actual classroom dates, adjusting for holidays and gaps.

One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

| Record | Teacher evidence |
| --- | --- |
| Learner code and date | |
| Concept and case used; prior exposure | |
| Explanation / trace / limits | |
| Support and first repair | |
| Changed case and delayed outcome | |
| Next teaching action | |

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

## Bridge to the next topic

Week 5 uses conditions in if elif and else branches, then tests exact boundaries.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

### ▼ Earlier wording, alternatives and picture transcripts

#### Plan the delayed return

#### Project evidence from this case

# Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

## Reference · Alternative planning checklist

> Use alongside the main route without repeating the teaching cycle.

### Planning and assessment reference

Teaching target: Translate stated requirements into **comparison**s

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

Before class, solve the supported practice: For (minutes, quiet) cases (11, True), (12, False), and (13, True), write the time **comparison** and final suitability values before running the program.

Reasoned practice answer: Time results are True, True, False. Combined results are True, False, False; the second case fails quiet and the third fails time.

Check every learner before discussion: Use the worked case for **Boolean** conditions. Proposed explanation: “At most excludes the boundary”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: At most 12 includes 12; compare < and <= at exactly 12.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Change age to 16. Require a **prediction** and a reason before comparing. Expected result: The result becomes False. Reason: The lower condition remains True, but the upper condition fails.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

| Criterion | Not yet | With support | Independent |
| --- | --- | --- | --- |
| Explanation | Names an answer without a deciding fact | Uses a hint to link fact and decision | Explains the deciding fact in own words |

| Criterion | Not yet | With support | Independent |
|---|---|---|---|
| Trace or test | Cannot connect input and result | Completes steps with prompts | Traces or tests a changed case accurately |
| Limits and revision | Claims more than evidence supports | Names a limit after a prompt | States a defensible limit and useful next test |

Project checkpoint: Python readiness portfolio. Keep runnable code, **prediction**s, actual **output**s, one repaired error and an explanation of a function on changed **input**s.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback and evidence records**

REFERENCE · SECTION 8

# Reference · Feedback and evidence records

> Keep individual first attempts separate from supported rehearsal.

## Feedback and repair

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

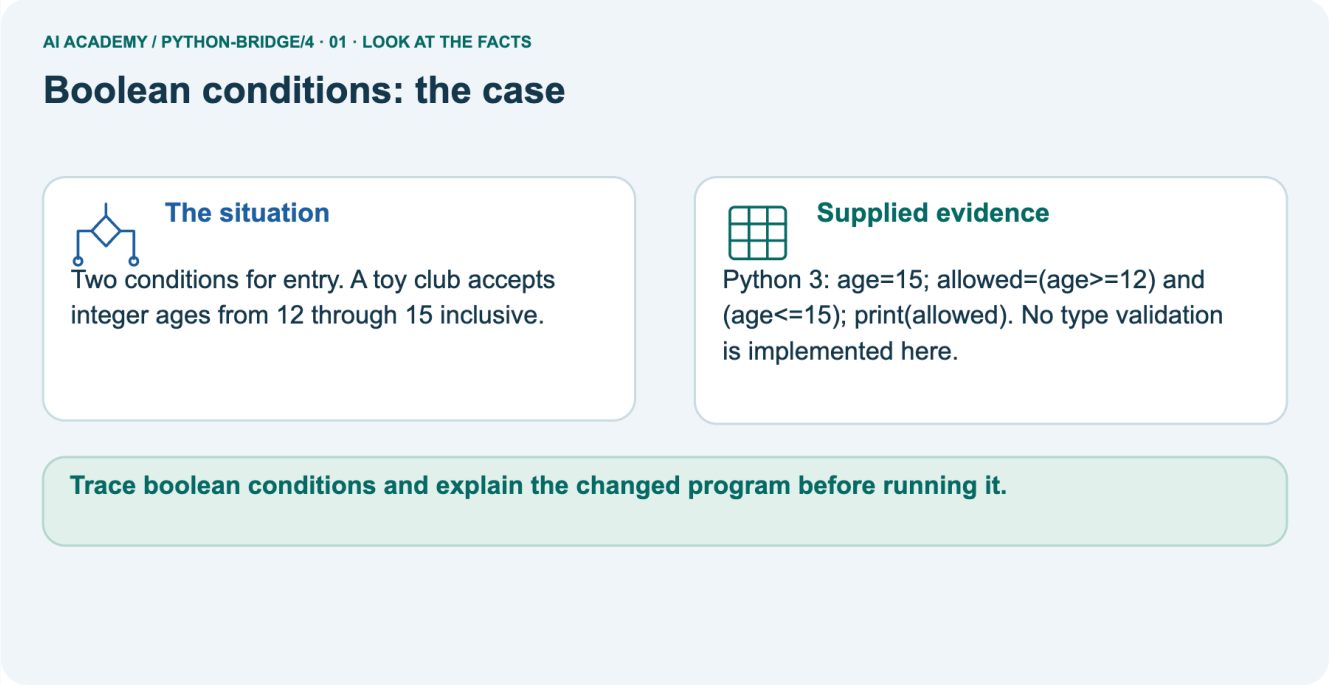
▼ Reference · Teach with the six pictures

REFERENCE · SECTION 9

# Reference · Teach with the six pictures

> Use the relevant picture for a specific reasoning move.

## 01. Boolean conditions: the case



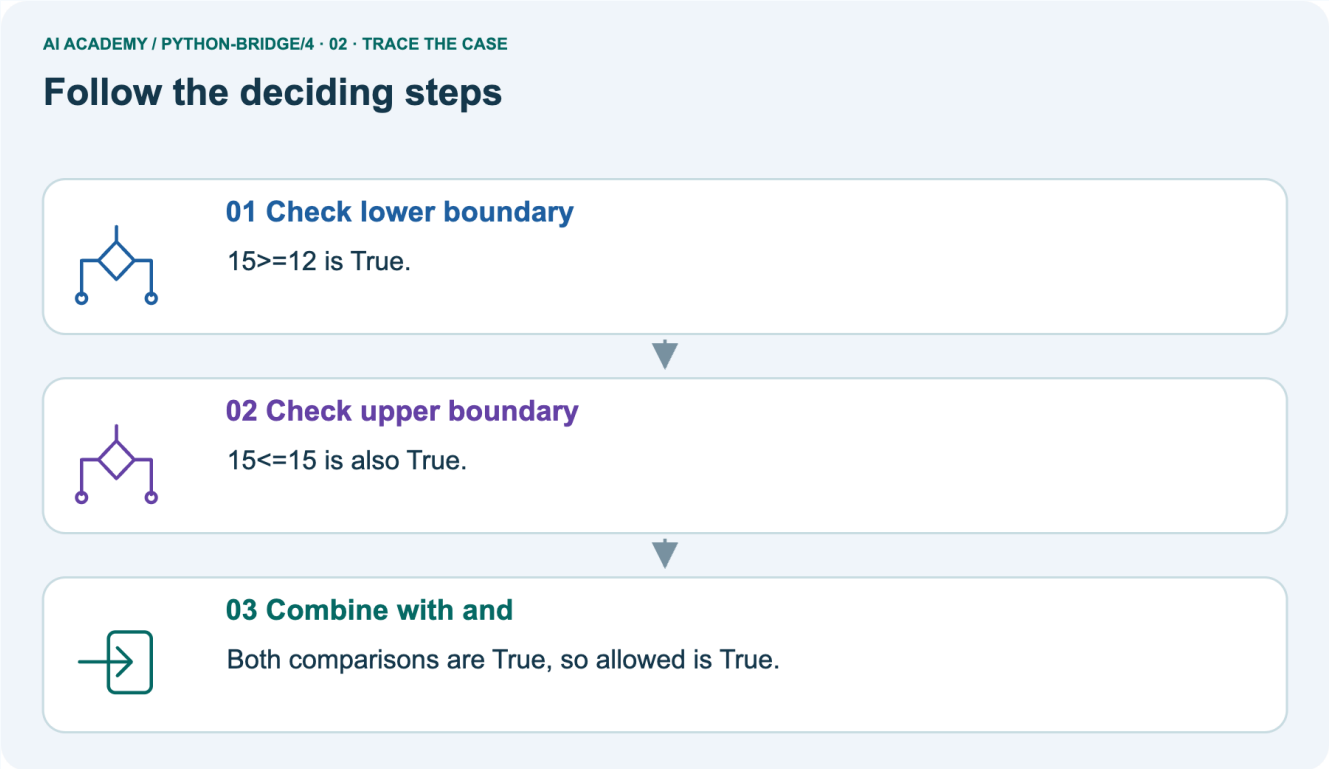

Follow the labelled picture with the matching explanation. Open the vector diagram

Two conditions for entry. A toy club accepts integer ages from 12 through 15 inclusive. Supplied facts: Python 3: age=15; allowed=(age>=12) and (age<=15); print(allowed). No type validation is implemented here.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

## 02. Follow the deciding steps



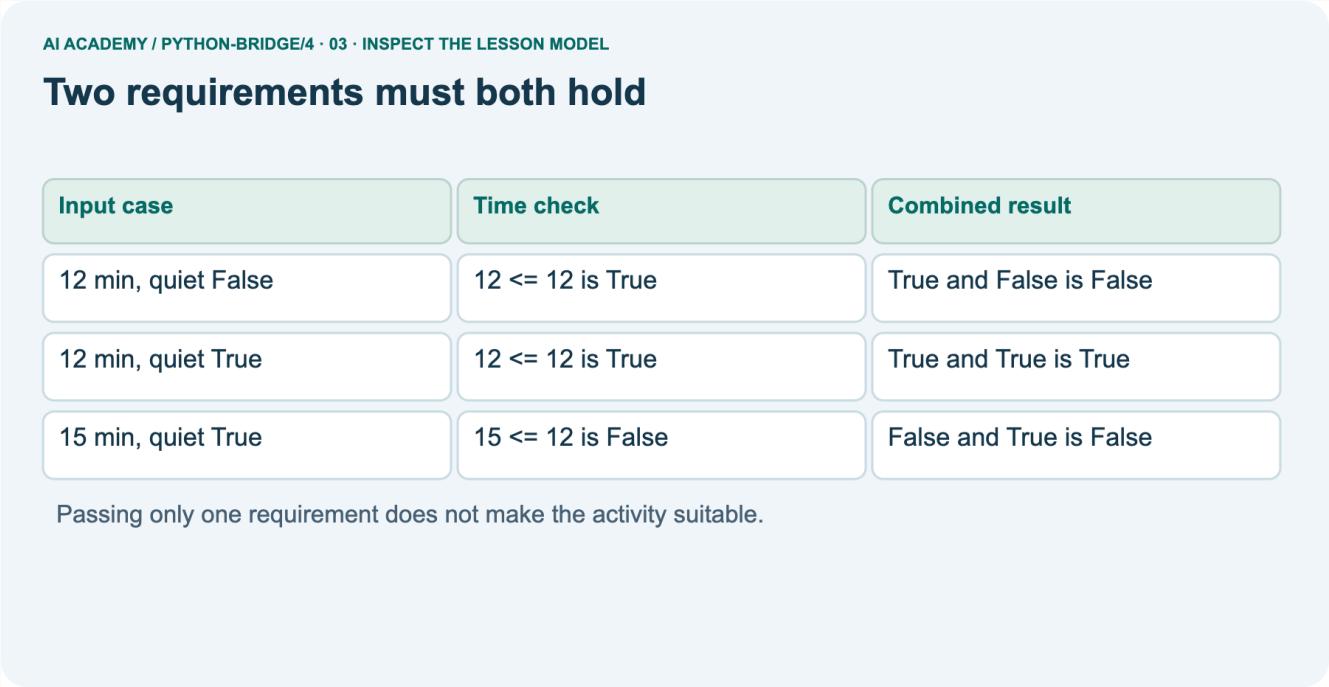

Follow the labelled picture with the matching explanation. Open the vector diagram

Check lower boundary: 15>=12 is True. → Check upper boundary: 15<=15 is also True. → Combine with and: Both **comparison**s are True, so allowed is True.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

## 03. Two requirements must both hold



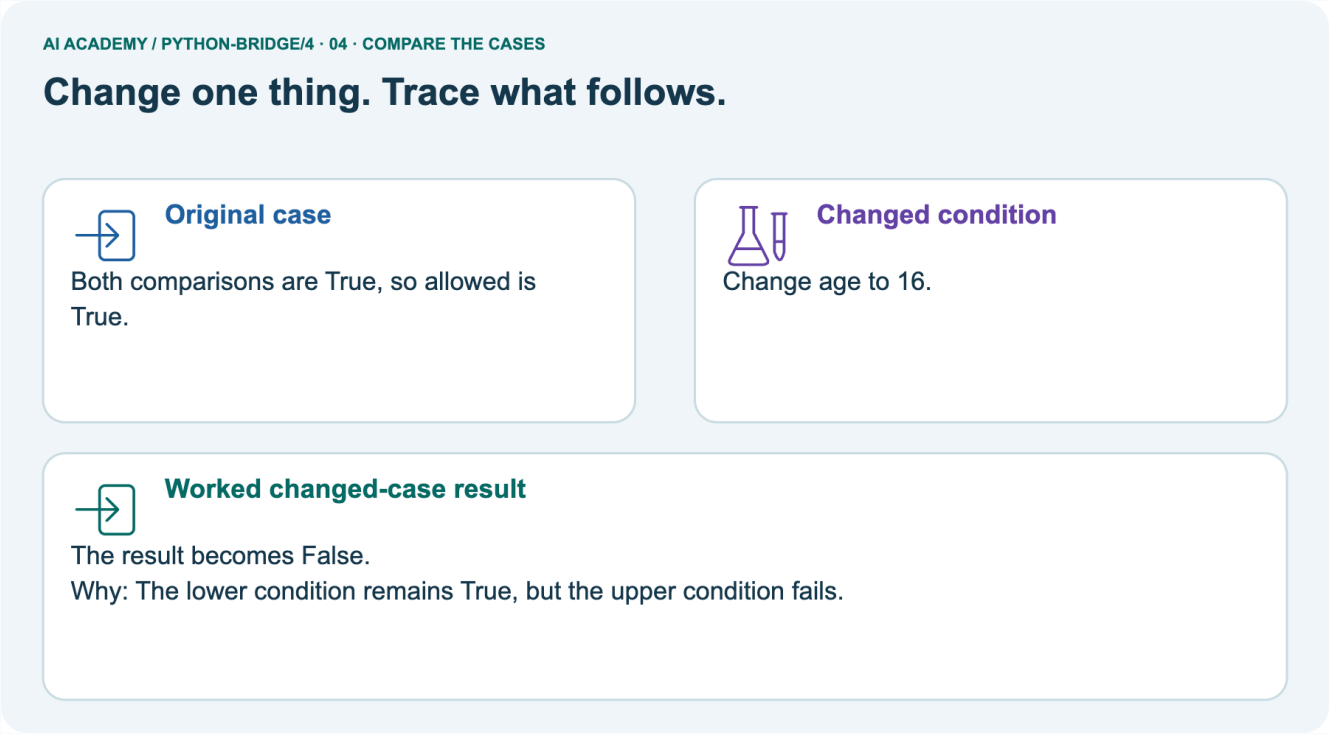

Follow the labelled picture with the matching explanation. Open the vector diagram

Two requirements must both hold. 12 min, quiet False | 12 <= 12 is True | True and False is False. 12 min, quiet True | 12 <= 12 is True | True and True is True. 15 min, quiet True | 15 <= 12 is False | False and True is False

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

## 04. Change one thing. Trace what follows.




Follow the labelled picture with the matching explanation. Open the vector diagram

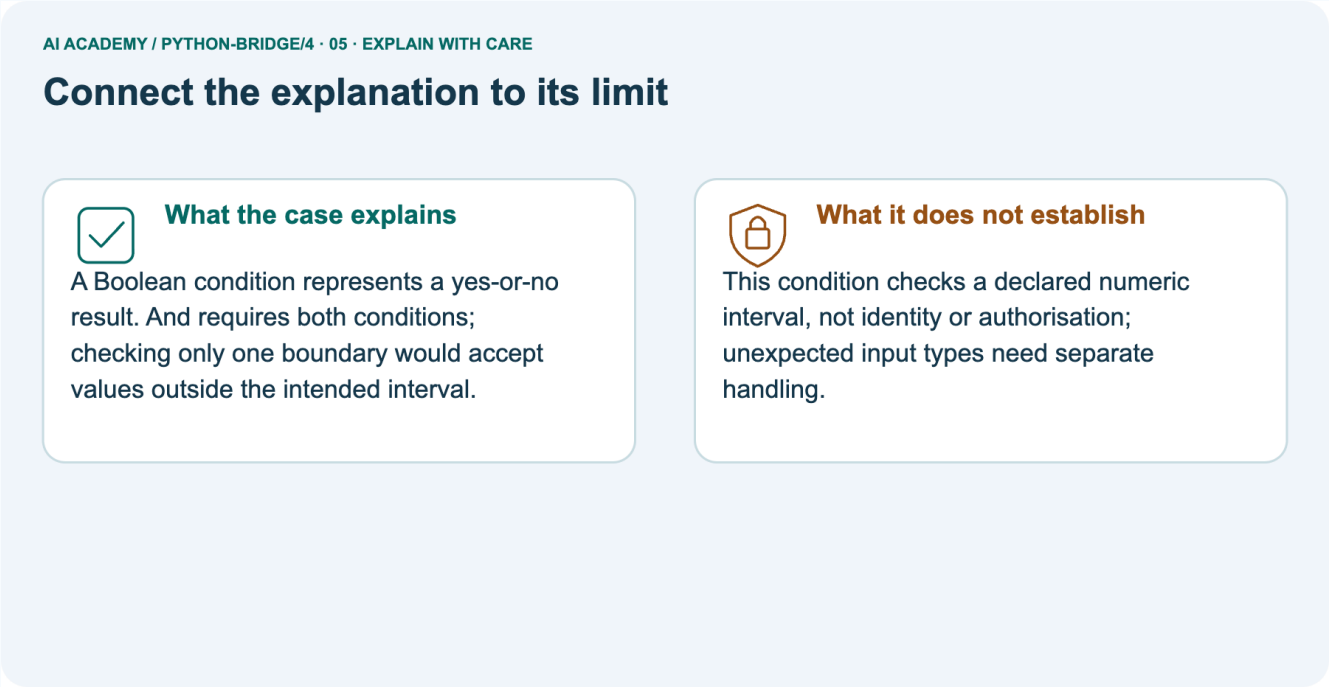
Original case: Both **comparison**s are True, so allowed is True. Changed condition: Change age to 16. Expected result: The result becomes False. Reason: The lower condition remains True, but the upper condition fails.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The lower condition remains True, but the upper condition fails.

## 05. Connect the explanation to its limit




Follow the labelled picture with the matching explanation. Open the vector diagram

A **Boolean** condition represents a yes-or-no result. And requires both conditions; checking only one boundary would accept values outside the intended interval.

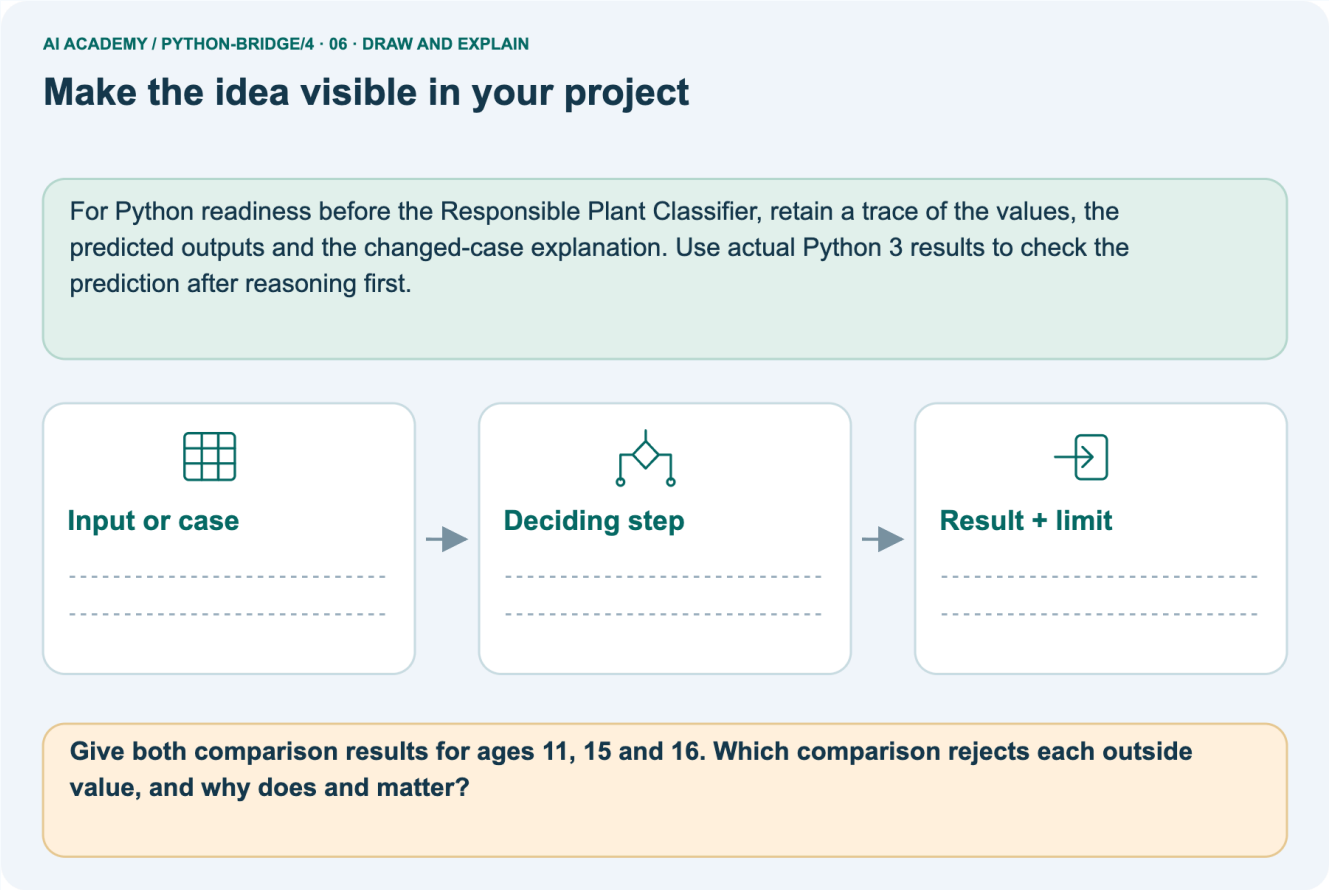
Limit: This condition checks a declared numeric interval, not identity or authorisation; unexpected **input** types need separate handling.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: This condition checks a declared numeric interval, not identity or authorisation; unexpected **input** types need separate handling.

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

Draw your **input**, deciding step, result and limit. Supported question: Give both **comparison** results for ages 11, 15 and 16. Which **comparison** rejects each outside value, and why does and matter?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Age 11 gives False/True, so False. Age 15 gives True/True, so True. Age 16 gives True/False, so False. Both interval boundaries must hold.

## ▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

### 01. Boolean conditions: the case

### 02. Follow the deciding steps

### 03. Two requirements must both hold

### 04. Change one thing. Trace what follows.

### 05. Connect the explanation to its limit

### 06. Make the idea visible in your project

## ▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.